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AI Context Exporter

Official Documentation

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Overview & Business Value AI Context Exporter is a comprehensive C++ solution designed for solo developers, indie creators, and AAA studios looking to standardize their Data-Driven Pipeline. Built with the philosophy that leveraging advanced AI shouldn't require a computer science degree, the tool translates Unreal Engine binary files (.uasset, .umap) into highly optimized, structured text formats (JSONL). It serves as the ultimate bridge between project architecture and Large Language Models (LLMs) or Agentic AI, empowering both non-programmers and seasoned engineers to build, audit, and scale their projects efficiently.

- **Enterprise-Grade Security & Air-Gapped Operation:** The tool operates 100% offline within the studio's intranet. There are no external dependencies, third-party APIs, or hidden telemetry. Proprietary code and project logic never leave the local hardware, ensuring strict compliance with corporate Non-Disclosure Agreements (NDAs).

- **Objective Code Audits & Architectural Health:** Beyond AI extraction, the engine functions as a technical debt scanner. It enables Technical Directors to perform Code Reviews on visual Blueprints using quantifiable metrics, saving dozens of hours in debugging massive projects.
- **API Cost Optimization (RAG Pipeline):** Generating AI context can become cost-prohibitive if "noise" or redundant engine data is exported. Through advanced compression and delta serialization, this tool massively reduces the payload size (JSONL), directly decreasing token consumption and billing costs on platforms like OpenAI, Anthropic, or Vector Database systems.
- **Zero Friction:** No Python environments to configure and no external dependencies. A drop-in installation that produces .jsonl files ready to be integrated into any workflow.

Architectural Health & Project Audits The core operational value of the tool lies in its ability to audit large-scale projects and pinpoint bottlenecks ahead of certification cycles:

- **Blueprint Node Complexity:** Objectively measures the quality of visual scripting. The system extracts the total number of nodes, maps execution links, calculates Hard Cast Dependencies, and generates a **Spaghetti Index Score**. This identifies unsustainable or performance-prone logic.
- **Technical Debt & Orphan Assets:** Silently scans C++ code and Blueprints for pending comments (TODO, FIXME, BUG, HACK), centralizing them into an actionable report. Additionally, it isolates empty or unreferenced assets that bloat package size.
- **Performance & Hierarchy:** Evaluates the Data-Driven Rate, warns about extreme Max Inheritance Depths, and identifies bottlenecks by listing all Actors with Tick enabled.

The Extraction Engine & Context Optimization The core process runs on an asynchronous background Worker, allowing the Unreal Engine editor to remain responsive even during massive exports.

- **Smart Delta Serialization:** Evaluates property values against their Class Default Object (CDO). The engine only exports variables explicitly modified by the developer. This purges native engine redundancy and

reduces the resulting file size by up to 80%, ensuring agile processing by the AI.

- **AST & Visual Logic Translation:** Parses uncompiled visual programming, translating execution pins, pure math logic, and variable sets into strictly structured JSON format.
- **Component Architecture Extraction (ECS):** By reading the CDO and the Simple Construction Script, the tool exposes the exact hierarchy and spatial relationships of component templates.
- **Path Compression:** Implements an aliasing system that substitutes deep package paths (/Game/...) with compressed integer identifiers, saving valuable tokens in the LLM's context window.

Integrated Multilingual Support The extraction engine natively supports 8 languages, dynamically adapting to the active Unreal Editor interface: English (en) (Native), Japanese (ja), Simplified Chinese (zh-Hans), Korean (ko), German (de), Spanish (es), French (fr), Brazilian Portuguese (pt-BR).

Data Architecture Note: To ensure strict compatibility with AI parsers and RAG systems, the output folder structure (e.g., 01_AI_Insights_And_Audits) is invariably generated in English. LLMs exhibit superior analytical performance when processing base structures in this language.

Extraction Modules (Targeted Scopes) The interface allows surgical control over the data payload. Toggle individual modules based on sprint requirements or the specific AI task:

- **Source Code (.cpp / .h):** Raw C++ implementation and Engine .ini configuration files.
- **Local Plugins:** Inclusion of local third-party plugin source code (subject to manual license validation and user EULA compliance).
- **Blueprints (Logic & Nodes):** Uncompiled logic, properties, execution links, and component structures.
- **World Levels (Actors & Partitions):** Actor transforms, localized properties, and tags within .umap physical environments.
- **VFX & Materials:** Niagara Systems and Material Instances.
- **Animations:** Animation Blueprints, Sequences, and Montages.
- **AI Logic:** Behavior Trees, Blackboards, and StateTrees.
- **Audio Sounds:** MetaSound graphs and SoundCue structures.
- **Data Assets:** Primary Data Assets, Data Tables, and String Tables.

- **Input Mappings:** Enhanced Input contexts and actions.
- **Physics:** Physical Materials and Props.

Interactive Analytics Dashboard Upon extraction completion, the system consolidates fragmented logs into a master ledger, presented via an analytical dashboard inside the editor:

- **Module Breakdown:** Graphical representation that breaks down physical disk size and file count by asset type, facilitating immediate identification of memory bloat.
- **Top 20 Rankings:** Automatically generated interactive lists exposing the most complex Blueprints, heaviest memory assets, and most referenced assets in the project.

Quick Start Guide

1. Launch your Unreal Engine project.
2. On the top menu bar, navigate to *Tools > Asset Context Exporter For AI*.
3. Select a strategic preset (e.g., *Core Architecture* or *Full Project*).
4. Under *Export Settings*, define your format (Monolithic JSONL for vector databases or Fragmented for standard context windows).
5. Define the output directory.
6. Click *RUN CONTEXT EXPORT*.
7. Once finished, click *Open Export Folder* to attach the telemetry to your selected AI model.

Marketplace Compliance The plugin incorporates an active safety mechanism (`bExcludeNoAIAssets`) designed to respect Epic Games FAB "NoAI" directives, automatically bypassing flagged assets during extraction. If company protocols require disabling this protection for strictly local processing, technical and legal responsibility falls entirely on the user's environment.

Technical Specifications

- **Supported Platforms:** Windows 64-bit
- **Supported Engine Versions:** 5.5, 5.6, 5.7
- **Support Contact:** mhvan_mx@outlook.com

